

Math 472: Homework 07

Due Monday, March 23

Problem 1 (Exercise 9.8). Let $Y_1, \dots, Y_n$ denote a random sample from a probability density function $f(y)$, which has unknown parameter θ . If $\hat{\theta}$ is an unbiased estimator of θ , then under very general conditions

$$\text{Var}(\hat{\theta}) \geq \frac{1}{\mathcal{I}(\theta)}, \quad \text{where} \quad \mathcal{I}(\theta) = n\mathbb{E} \left[-\frac{\partial^2}{\partial \theta^2} \left[\log(f(Y)) \right] \right].$$

(Note that in the definition of $\mathcal{I}(\theta)$, the Y is a random variable with density function f .) If $\text{Var}(\hat{\theta}) = \frac{1}{\mathcal{I}(\theta)}$, the estimator $\hat{\theta}$ is said to be *efficient*.

Suppose that $p(y)$ is the Poisson probability function with mean $\lambda > 0$. Show that $\bar{Y}$ is an efficient estimator of λ .

Problem 2. Given a function $h : \mathbb{R} \rightarrow \mathbb{R}$, we say that $x^* \in \mathbb{R}$ is a *maximizer* of h if

$$h(x) \leq h(x^*) \text{ for all } x \in \mathbb{R}.$$

We say that $x^* \in \mathbb{R}$ is a *local maximum* of h if there exists some $\delta > 0$ such that

$$h(x) \leq h(x^*) \text{ for all } x \in (x^* - \delta, x^* + \delta).$$

We say that a function h is *strictly increasing* if

$$h(x) < h(y) \text{ whenever } x < y.$$

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function, and let $g = \phi \circ f$, where ϕ is a strictly increasing function whose domain contains the range of f .

- (a) Show that x^* is a maximizer of f if and only if it is a maximizer of g . (You don't need to assume that f or ϕ is differentiable.)
- (b) Show that x^* is a local maximum of f if and only if it is a local maximum of g . Conclude that f and g have the same local maxima. (Again, you don't need to assume that f or ϕ is differentiable).
- (c) Does the same conclusion hold for local minima? [Hint: consider the function $-f(x)$.]

In our class, when finding the maximum likelihood estimator, we need to maximize the likelihood $L(\theta)$. The previous problem shows that it is enough to maximize $\log L(\theta)$, since $\phi(x) := \log(x)$ is strictly increasing. In fact, we can obtain stronger conclusions if we are allowed to assume that f and ϕ are differentiable, as shown in the next problem.

Problem 3. Let f, g , and ϕ be defined as in Problem 2. Further, assume that f and ϕ are both twice differentiable, and that $\phi'(x) > 0$ for all $x \in \mathbb{R}$.

- (a) Show that $x^* \in \mathbb{R}$ is a critical point of f if and only if it is a critical point of g .
- (b) Suppose that $x^* \in \mathbb{R}$ is a critical point of f and g . Show that $f''(x^*)$ and $g''(x^*)$ are either (i) both positive, (ii) both negative, or (iii) both zero.
- (c) What additional information does part (b) tell you about functions f and g ?

Problem 4. There is a subtlety to the assumptions used in Problems 2 and 3, which we highlight in this problem. Let $\phi : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function.

- (a) Show that if $\phi'(x) > 0$ for all $x \in \mathbb{R}$, then ϕ is strictly increasing. [Hint: use the mean value theorem].
- (b) Show that the converse to part (a) does not hold (i.e., give a counterexample).

Problem 5. Write R code to replicate Figure 9.2 (page 449) in the textbook.

Problem 6 (Exercise 9.3 and 9.15?). Let $Y_1, \dots, Y_n$ denote a random sample from the uniform distribution on the interval $(\theta, \theta + 1)$. Let

$$\hat{\theta}_1 = \bar{Y} - \frac{1}{2} \quad \text{and} \quad \hat{\theta}_2 = \max(Y_1, \dots, Y_n) - \frac{n}{n+1}.$$

- (a) Show that both $\hat{\theta}_1$ and $\hat{\theta}_2$ are unbiased estimators of θ .
- (b) Find the efficiency of $\hat{\theta}_1$ relative to $\hat{\theta}_2$.
- (c) Show that both $\hat{\theta}_1$ and $\hat{\theta}_2$ are consistent estimators of θ .

Problem 7. Let X be a random variable with cdf

$$F(x) = \frac{1}{2} + \frac{1}{\pi} \arctan(x), \quad (x \in \mathbb{R})$$

(This is called a Cauchy random variable.)

- (a) Show that $\mathbb{E}[X]$ is not defined.
- (b) Let $Y_n = \frac{X_1 + \dots + X_n}{n}$. For $n \in \{100, 10000, 100000\}$, sample a large number of independent replicates of Y_n , and plot the histograms. Explain what you see.

Problem 8 (Exercise 9.37). Let $X_1, \dots, X_n$ denote n independent and identically distributed Bernoulli random variables such that

$$\mathbb{P}[X_i = 1] = p \quad \text{and} \quad \mathbb{P}[X_i = 0] = 1 - p$$

for each $i = 1, \dots, n$. Let $S_n = X_1 + \dots + X_n$. Show that S_n is sufficient for p by using the factorization criterion.

Problem 9. Suppose $Y_1, \dots, Y_n$ is a random sample from a geometric distribution with parameter p . Show that $\bar{Y}$ is sufficient for p .

Problem 10. Let $Y_1, \dots, Y_n$ be a random sample from a normal distribution with mean μ and variance σ^2 .

- (a) If μ is unknown and σ^2 is known, show that $\bar{Y}$ is sufficient for μ .
- (b) If μ is known and σ^2 is unknown, show that $\sum_{i=1}^n (Y_i - \mu)^2$ is sufficient for σ^2 .

Problem 11. Suppose $Y_1, \dots, Y_n$ is a random sample from a Poisson distribution with mean $\lambda > 0$.

- (a) Use the factorization criterion to find a sufficient statistic for λ .
- (b) Find the MLE $\hat{\lambda}$ of θ .

Problem 12. Suppose $Y_1, \dots, Y_n$ is a random sample from a uniform distribution with pdf

$$f(y | \theta) = \frac{1}{2\theta + 1} \cdot \mathbf{1}_{[0 \leq y \leq 2\theta + 1]}.$$

Find the mle of θ .

Problem 13 (Optional). Make a math meme related to statistics.